

Holographic Storage for the Cloud: advances and challenges

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Holographic Storage is an old idea that has always promised high density and fast random access, but has never been commercially competitive with Hard Disk Drives (HDDs) and Solid State Devices (SSDs). In Project HSD at Microsoft Research we asked the question: “Does holographic storage finally make sense for cloud storage?” This article describes our journey toward answering this question.

We achieved $1.8\times$ higher density than the previous state-of-the-art, using commodity components available today and leveraging machine learning to compensate for the noise and distortions introduced by commodity components. This uncovered two new challenges which are the focus of this article: achieving high end-to-end energy efficiency without sacrificing capacity, and spatial multiplexing without mechanical movement.

Improving end-to-end energy efficiency requires joint optimization across low-level media parameters and higher-level system parameters that govern background maintenance operations such as read refresh and

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garbage collection. We developed new physics models of the media; analytic and simulation models of the media access and background media maintenance; and workload-driven optimization to find optimal parameter combinations. These techniques resulted in a 14× improvement over the previous approach for typical workloads without sacrificing capacity. We also designed the first scalable and mechanical movement free spatial multiplexing system for holographic storage.

Despite these advances, we conclude that currently, holographic storage is still far from the combination of density, capacity scaling, and energy efficiency needed to compete with the incumbent technologies. We need fundamental advances in the physical media that improve energy efficiency by another 1–2 orders of magnitude without reducing data density. Further advances in optics are also required to achieve spatial multiplexing that is simultaneously scalable, low-loss, and high-density.

CCS Concepts: • **Hardware** → **Emerging optical and photonic technologies** • **Computer systems organization** → **Cloud computing**;

Additional Key Words and Phrases: Holographic storage, read erasure, write erasure, garbage collection, end-to-end simulation

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1 Introduction

Cloud storage is going through an inflection. For warm rewritable storage in the cloud this has driven exponential demand for magnetic **hard disk drive (HDD)** capacity. This demand is already at exabyte scale and will reach zettabyte scale in the mid-2020s [81], even sooner than previously predicted [21]. Faced with exponentially increasing demand, cloud providers such as Microsoft have relied on corresponding growth in hardware GB/\$ to keep costs from exploding. This has usually meant increased capacity per device at a more-or-less constant dollar cost per device. However, we can no longer rely on this, and if we do not fundamentally innovate in storage hardware then the demand will outstrip the supply and the cost of cloud storage will increase dramatically.

There has been so much innovation in the last decade in computing and networking, much of it driven by the needs and scale of the cloud. Yet, we have seen little fundamental innovation in storage. HDDs and SSDs are no longer improving exponentially in GB/\$: they are in the mature phase of their the technology “S-curve” [80] shown in Figure 1. While GB/\$ continues to improve for these technologies, the rate of improvement is slowing down. It is not clear yet whether new technologies such as HAMR will start a new S-curve for HDD density. However, for HDDs in particular, an even bigger challenge is that the access rates (IOPS) have been flat for decades. Many workloads cannot benefit from improvements in device capacity as they do not come with a corresponding increase in device IOPS.

Our focus in this article is on warm storage, i.e., rewritable storage with a relatively high access rate requirement. Other storage tiers such as archival storage are facing similar challenges. The incumbent technology — magnetic tape — is also mature and has only been improving slowly. DNA storage [9, 18, 32] and more recently storage in glass [3, 4] are two examples of disruptive new technologies that are being developed to replace magnetic tape. However, these technologies will not meet the needs of warm data storage. They are not rewritable and they do not provide sufficiently high access rates particularly for small random accesses. On the other hand, they offer a level of durability, density, and low cost that is beyond what is needed for warm data.

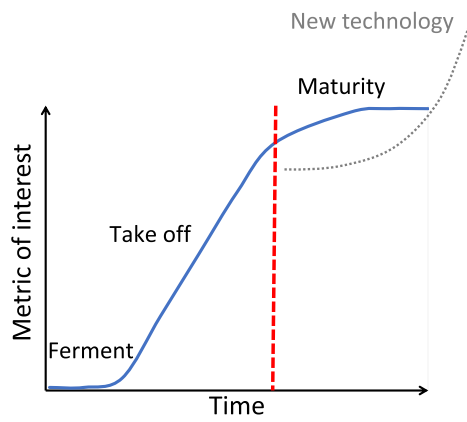


Fig. 1. Technology S-curve.

Warm storage is thus ready for disruption by new storage media. Cloud storage also offers a unique opportunity to innovate in overall storage design. HDDs and SSDs initially relied on consumer device demand to drive scale and volume: the effect is that even in the cloud, they are constrained to use legacy form factors and interfaces, e.g., the 3.5" form factor for HDDs. Cloud storage (like computing and networking) by itself is already at a scale that justifies revisiting these design decisions. Cloud services are designed at rack scale, and so we can design storage hardware for the form factor that makes the most sense for the technology, at least up to rack scale. Both hardware and software details are abstracted away from the user, who sees only high-level service APIs. This frees us to rethink the software and hardware interfaces and the storage stack.

Why holographic storage? Holographic storage [25] records data as holograms in a photosensitive medium. It is an old idea with an extensive research literature. It has several attractive properties for warm data storage: it can use a rewritable, low-cost, bulk 3D medium. Unlike flash, this does not wear out — it can sustain an indefinite number of program/erase cycles. It uses optics to read and write data, which means it can provide high access rates and low latency. Unlike magnetic media (HDDs), CMOS-based memories (e.g., flash SSDs), or 2-D optical storage (e.g., DVDs), rewritable holographic storage can be truly 3-dimensional. A single volume (or “zone”) can in theory store thousands of holograms, each of which encodes tens of kilobytes to megabytes of data. The optical read/write mechanisms can be decoupled from the bulk media, compared to the tight integration required, e.g., with HDD heads which must be in extremely close proximity to the medium. Holograms can be randomly accessed for read within each zone. Holograms are read and written as entire 2-D images each encoding a large number of symbols, again unlike magnetic media where a read/write head can only read a single symbol at a time. These properties make holographic storage attractive for durability, density, cost, throughput and IOPS.

However despite these attractive properties, holographic storage has never been successfully commercialized. Even in the research literature, there are few examples of end-to-end systems or prototypes for **Write Once Read Many (WORM)** holographic storage; even fewer of rewritable holographic storage; and none approaching the capacity, scale, or performance that would be needed for commercial viability. In Project HSD [64] we asked the question “does rewritable holographic storage finally make sense in the cloud era”? We note that advances in optical technology has enabled high storage densities as well as performance that is orders of magnitude better than possible in the past. At the same time, commoditization of these technologies has driven down

the cost of the components. Are these trends, combined with the freedom to redesign the storage stack from top to bottom, sufficient to make holographic storage viable for the cloud?

Learnings from Project HSD. For the previous wave of research in holographic storage (until the early 2000s), achieving high density at low cost was a challenge. The key optical technologies – cameras, SLMs (Spatial Light Modulators), and lasers were low-resolution and/or high-cost. In Project HSD we demonstrated that high density is achievable using commodity components: we have recorded densities [13] $1.8\times$ higher than the previous state-of-the-art [15]. To do this we leveraged the commoditization and improvement of optical components in the last two decades. We used machine learning to compensate for the distortions added by commodity optics [50] and improved the state-of-the-art in measurement and modeling of holographic media [23].

This uncovered two new challenges that are the focus of this article. First, the end-to-end energy efficiency of rewritable holographic storage was 2–3 orders of magnitude too low to be practical. While there has been previous research on minimizing the optical energy used for individual reads and writes, the challenge is to achieve *end-to-end* energy efficiency *without sacrificing net capacity*. This means we must consider the overhead caused by media maintenance operations such as read refresh and garbage collection that are essential to cope with *write erasure* and *read erasure*. In this article we describe, model, and optimize end-to-end energy efficiency. Compared to the previous state-of-the-art for optimizing media energy use [57], this improved end-to-end efficiency by $14\times$ for typical workloads without sacrificing capacity.

Second, we identify optical spatial multiplexing as a key challenge. Multiplexing the camera and SLM over a large amount of data is essential for achieving competitive cost per GB. This problem is not well addressed in previous research, with most work assuming physical movement of the medium. This fundamentally limits IOPS making the technology less attractive than HDDs (which already suffer from low IOPS). Thus the challenge is to achieve spatial multiplexing without requiring mechanical movement. We describe in this article a mechanical movement free design for spatial multiplexing.

However, neither challenge has been fully solved. Energy efficiency is still a barrier to commercial viability, and we need another 1–2 orders of magnitude improvement on this metric. We believe this is only possible with fundamental advances in holographic materials. Spatial multiplexing also continues to be a challenge. Our spatial multiplexing design can either achieve good power efficiency or high density, but not both simultaneously. Further advances in optics are needed to overcome this challenge.

2 Background

2.1 Incumbent Technologies: HDDs and SSDs

Figure 2 shows the historical capacity per device of HDDs. Until around 2010 this was a true Moore’s Law exponential driven by shrinking feature sizes. However, since then the rate has slowed as improvement is driven by increasingly complex and “one-off” advances such as helium-filling to increase platter count; shingling to increase the number of tracks; and most recently energy-assisted technologies such as HAMR [65]. It is not yet clear whether HAMR will allow significant further scaling. It is a very complex technology and despite more than a decade of research, has not yet deployed commercially at scale [69]. Due to this complexity it is also likely to increase per-device cost when it ships.

The real challenge with magnetic HDDs however is that they are less and less able to sustain the access rates (IOPS/TB) required by warm data workloads. A viable technology for cloud storage needs to provide an IOPS/TB that is equal or higher than the IOPS/TB of the workload. However, HDD IOPS per device have stayed almost completely flat for several decades, which means that

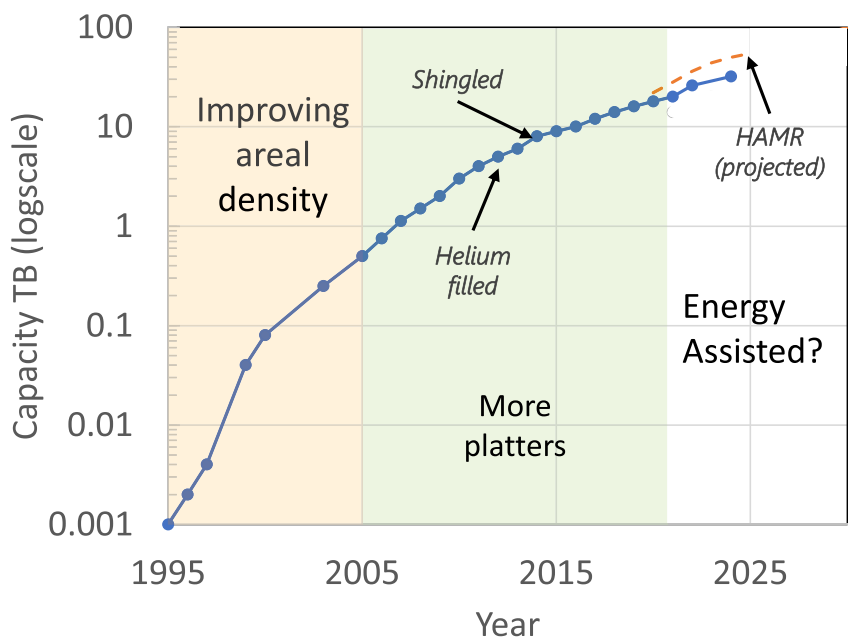


Fig. 2. HDD per-device capacity over time.

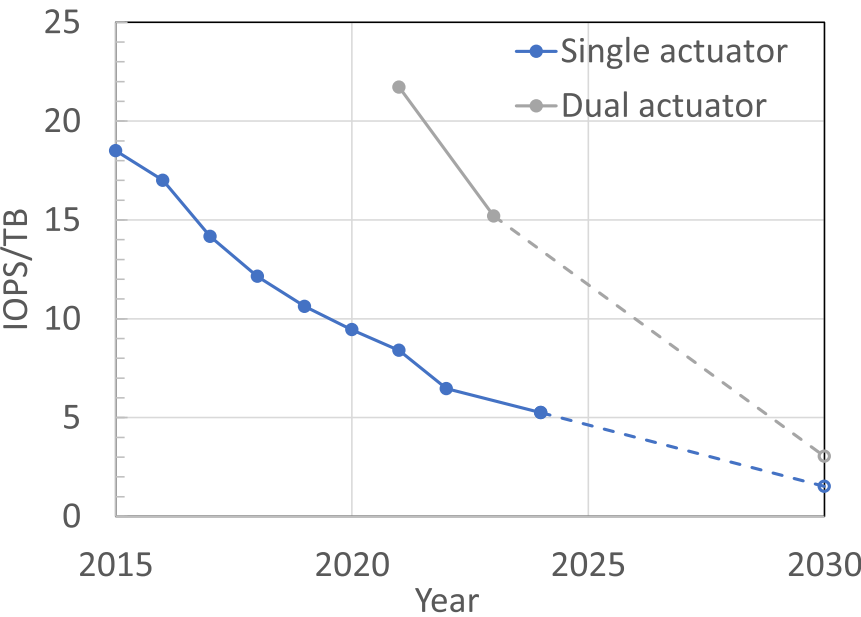


Fig. 3. HDD IOPS/TB over time.

every increase in per-device capacity comes with a corresponding drop in IOPS/TB. Figure 3 shows the recent history of HDD IOPS/TB projected out to 2030. Recently announced dual-actuator HDDs give a one-off improvement in IOPS/TB but this comes at a price premium and is not a scalable solution: the number of actuators cannot simply be increased with each new generation of devices.

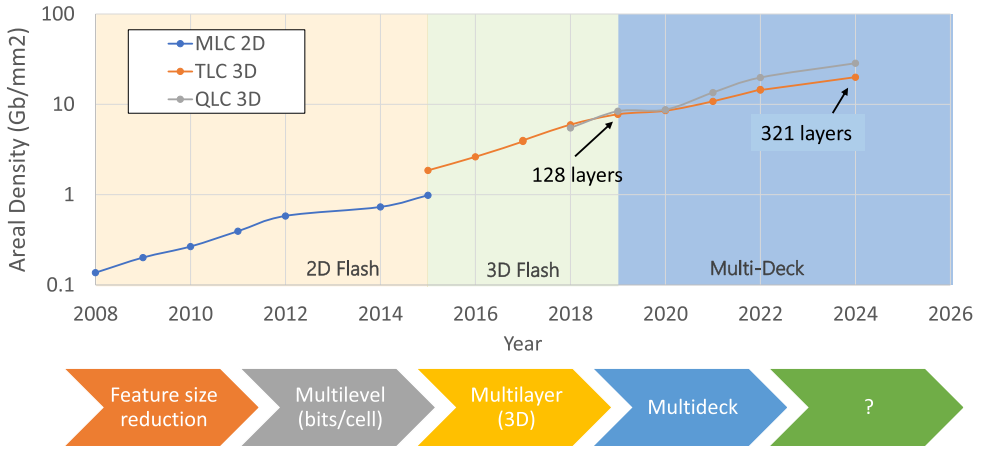


Fig. 4. NAND flash chip areal density over time.

The increasing use of **shingled magnetic recording (SMR)** for high capacity is another challenge – it reduces effective IOPS/TB (not shown in the figure) for random write workloads due to write amplification [2, 67].

The decline in IOPS/TB results in *stranded capacity* which is a huge problem for cloud storage providers. If the average IOPS/TB of the workload is higher than that of the device, then we cannot use the full capacity of the device, and thus the increase in per-device capacity gives no benefit. In some cases stranded capacity can be resold as “low-latency archival storage” [59] – however this is a relatively small gain in revenue as it must compete on cost with tape and other archival technologies. The only other solution to stranded capacity is to move more of the workload onto SSDs at significantly higher cost/GB.

Will SSDs eventually become cheap enough to solve this problem? Historically SSD costs have been driven by the increasing areal density of NAND flash chips. Figure 4 shows this historical areal density. For 2D flash, this effectively stopped increasing in 2010 and further increases were achieved by increasing the number of 3D flash layers. Scaling in the vertical dimension gives only linear benefits (compared to quadratic for 2D features size reduction) and is also limited by the aspect ratio achievable with etching and deposition [71, 77]. We may have already hit the limits of this – recent advances have relied on a multideck approach (also known as “string stacking”) that stacks individual devices on top of each other. While this increases the density and amortizes some costs at the package and device level, it does not fundamentally reduce the per-bit cost at the chip level.

The only other option to increase bit density in NAND flash is to increase the number of bits per cell. TLC (3 bits/cell) is widely deployed in the cloud today – for some workloads QLC (4 bits/cell) can be used. However, this only gives a 33% improvement in capacity, is harder to manufacture, and has much lower ($\sim 4\times$) endurance which is challenging for write-heavy workloads. PLC (5 bits/cell) has been announced but not yet shipped, and the capacity improvement here is even smaller (25% over QLC). The fundamental challenge with this approach is that at each step the number of programmable levels per cell must be doubled but the gain is only one bit, hence the relative capacity gain shrinks with each generation.

There is thus an increasingly large gap between SSDs and HDDs. It is possible that an entirely new memory technology will appear to fill this niche – at the moment though there seems to be

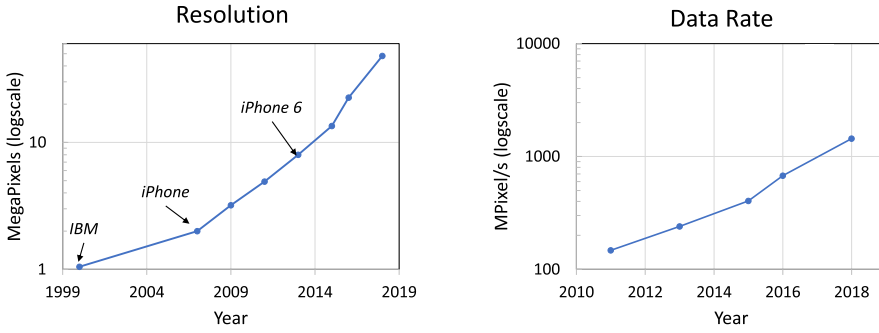


Fig. 5. Improvements in digital camera technology 1999–2018.

no viable candidate. Resistive memories [84], despite decades of research have not fulfilled their promise. The most successful technology in this space — Optane (3D Xpoint) — was unable to displace flash and has been discontinued [24]. The question that motivated this work was: can holographic storage fill this gap?

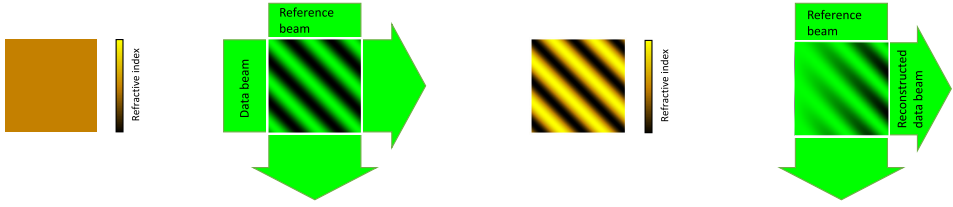
2.2 Holographic Storage

Holography was discovered in 1948, but needed the development of the laser (discovered in 1960) to be used in practical applications. The first holographic storage system was demonstrated in 1963 [49]. Since then there has been an extensive literature on different aspects of the optics and physics of holographic storage [25], and a few attempts to build *rewritable* storage devices starting from the 70s [33, 39, 54] — the IBM system [14], designed in the late 90s, to our knowledge represents the most complete end-to-end data storage so far, and the highest previously reported density for rewritable holographic storage. However, this was never commercialized: in the early 2000s there were still several commercial and technological obstacles that would have made production and deployment at scale very difficult. HDDs were still in the exponential growth phase of their capacity curves making it very difficult for a new technology at the initial stages of the S-curve, to compete on cost/capacity. This we believe has now changed; and for cloud storage in particular the declining IOPS/TB of HDDs is a huge driver to find alternative technologies with higher IOPS/TB but that are still cheaper than flash SSDs. The ability to design at rack scale for the cloud, rather than 3.5" form factors also unblocks the use of commodity optical components.

Perhaps most importantly, the optical technologies that are key to holography have improved dramatically in the last two decades. Cameras in particular have become commodity devices with Moore's Law like improvements driven by the smartphone industry (Figure 5). Similarly commoditization of SLMs which are essentially miniaturized projection devices has been driven by advances in heads-up displays, projectors, and LIDAR. Laser quality and power as well as miniaturization have been driven by extensive use in optical communications as well as consumer devices. The components that were expensive, custom devices in 2000 are now available as commodity devices with higher resolution, speed, and quality at lower power and cost.

3 Holographic Storage Design

The core principle of holographic data storage is the recording of an optical interference pattern (a *hologram*) within a photorefractive material. Figure 6 shows hologram formation and readout in 2 dimensions.¹ An interference pattern is created by the intersection of two coherent laser beams



(a) Blank medium has uniform refractive index. (b) Interference pattern creates non-uniformity. (c) Non-uniformity persists as a Bragg grating. (d) Bragg grating diffracts reference beam to recreate data beam.

Fig. 6. Creating a persistent interference pattern.

within the medium (Figure 6(b)). This pattern results in local changes in refractive index, which persist after the light is removed (Figure 6(c)). To exploit this for storage, one of the beams is *modulated* with a 2D image that encodes a block (page) of data. This data beam is intersected with a reference beam to create the hologram. Once the hologram is written, the reference beam alone is used for readout — it diffracts off the hologram to recreate a noisy and attenuated version of the data beam from which the data can be decoded (Figure 6(d)). This kind of diffraction is known as Bragg diffraction [12] and the holograms are sometimes referred to as Bragg gratings.

Only certain materials have the right combination of properties to record holograms. The medium must be *photosensitive*, i.e., respond to light by changing its local optical properties (e.g., the refractive index) as a function of light intensity. One popular mechanism is for the light energy to cause a chemical change (such as polymerization) changing its refractive index. It fits well for WORM holographic storage but it is too slow, energy-intensive, and difficult to reverse to be useful for rewritable storage with high access rates.

For rewritable storage, the best known materials are electro-optical ones. Incident light on such materials results in a locally non-uniform electric field, which in turn causes a non-uniform refractive index, i.e., a Bragg grating. The most promising — and most studied — electro-optical holographic material is iron-doped Lithium Niobate ($\text{Fe}:\text{LiNbO}_3$) [25], which has good optical quality of holograms, speed of hologram formation, and density of holograms. It is manufactured in bulk using the Czochralski method [79] that is also used to manufacture semiconductor grade silicon, and thus has the potential for large-scale production. It is also very stable, and can be read and written to for decades without degradation; this is unlike, for example, flash memory which has a limited number of write cycles. It also has high durability, i.e., the rate of *dark decay* (the degradation of holograms over time without external action) is very low: a hologram's half-life is counted in years [38]. In this article we focus on iron-doped Lithium Niobate crystal, which we will refer to as the medium or simply the crystal.

This material is a crystal lattice of Lithium, Niobium, and Oxygen atoms with iron impurities (Fe^{2+} and Fe^{3+} ions) as a dopant material. Lithium Niobate has the required electro-optical properties and the Fe ions act as *electron traps* (Figure 7. Light at 532 nm wavelength (green) excites electrons from Fe^{2+} ions to the conduction band where they can diffuse freely. As this excitation happens more when the intensity of light is high, the net effect under non-uniform illumination is that electrons diffuse from the brightly lit areas to the darker areas. Some of the electrons moved to the dark areas of the interference pattern fall into these traps, converting Fe^{3+} to Fe^{2+} . This

¹This is simplified for illustration purposes. In reality holographic storage is 3-dimensional, and also the interference pattern is non-uniform as it encodes the data.

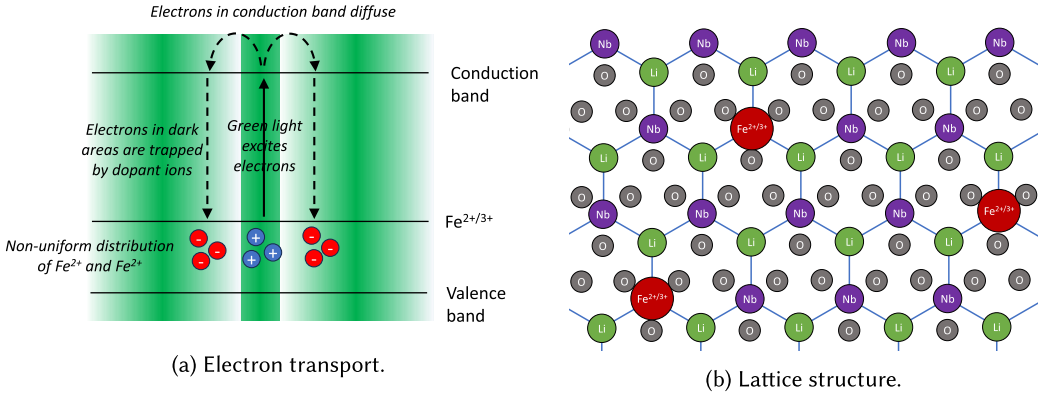


Fig. 7. Physics of rewritable holographic storage in Fe:LiNbO_3 .

results in a persistent change in electron density even after the incident light is removed, which in turn causes a local change in refractive index. Iron-doped Lithium Niobate is typically used with 532 nm laser light as it has maximum photosensitivity at this wavelength. As the read wavelength must match the write wavelength, the same laser source is used for both operations. Reads also result in redistribution of charge; as reading uses a single beam the illumination is uniform and hence the redistribution is toward a uniform charge distribution. Due to the lower power of reads and the lack of interference patterns that create strong peaks, this effect is much weaker than that of writes, but must still be accounted for cumulatively over thousands of read operations.

3.1 Key Concepts

Diffraction efficiency. The diffraction efficiency of a hologram is the ratio of light intensity diffracted into the readout beam to the light intensity of the reference beam. Typical diffraction efficiencies range from ranges from 10^{-5} to 10^{-3} . This means most of the reference beam power is not diffracted by the hologram into the readout beam, but simply passes through the crystal. Increasing diffraction efficiency of individual holograms makes reads more energy-efficient; however this requires stronger holograms which require more energy to write. Stronger holograms also means fewer holograms can be superposed within the same volume, reducing density.

Zoned storage. Holograms once written cannot be overwritten, but new holograms can be superposed on existing ones. All the superposed holograms can be simultaneously erased using lower-wavelength (UV) light, generated from low-cost UV LEDs. In a practical system, a larger volume of crystal would be subdivided into many *zones*, each of which can be independently erased. This is very similar to the zoned storage model [85] of Flash SSDs and shingled (SMR) hard disk drives. Pages can be appended to a zone, individual pages within a zone can be randomly access for read, and the entire zone can be erased (reset). To present the abstraction of rewritable storage, zoned storage devices must implement *garbage collection* to move live data from a zone that is about to be erased into a fresh zone.

Superposition of holograms. Individual holograms can encode kilobytes to megabytes of data within a relatively small volume of crystal, i.e., a zone. To achieve high density we need to superpose 100s to 1000s of hologram within each zone. This is done by changing the reference beam — for each unique reference beam, a different page can be written to the same zone, and these pages can be individually addressed by using the same unique reference beam. For this reason the

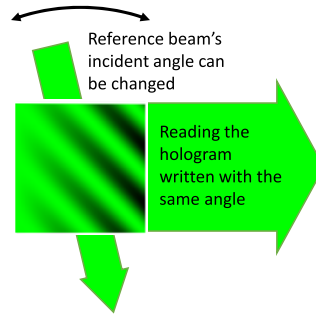


Fig. 8. Angular multiplexing.

reference beam is also referred to as the *address beam*. For this to work well, the different reference beams must be orthogonal — i.e., they must have high correlation with their own pages and low correlation with any other pages, regardless of the data encoded in the pages.

Multiple techniques exist for multiplexing within a zone. *Angular multiplexing* (Figure 8 changes the wavefront angle of the reference beam using, e.g., MEMS mirrors and lenses. This is known to give very high orthogonality through Bragg selectivity [12] and has been used successfully to write up to 5,000 holograms in a single zone [55]. *Phase multiplexing* uses **spatial light modulators** (SLMs) to encode one of a set of orthogonal phase codes on the reference beam [8, 27, 29, 30, 58]. *Wavelength multiplexing* uses different wavelengths of light for different pages [49, 62, 83]. In this article we focus on angular multiplexing, which is the most studied and most promising technique for high-density holographic storage.

In principle, the number of holograms that can be superposed with angular multiplexing is only limited by the available optical angle and the wavelength of the light. With a typical setup, this is around 3,000 holograms. In reality however, the hologram density is limited by the medium, which has limited dynamic range and also suffers from both write and read erasure. A common approximation is that the diffraction efficiency is inversely proportional to the square of the number of holograms superposed. However, this is a simplification — the actual relationship is more complex and depends on the write and read profiles used for each hologram. These profiles — as well as media maintenance operations such as garbage collection and read refresh — govern a complex tradeoff between density and energy efficiency.

Write erasure. Each superposed hologram slightly degrades the previously written holograms and decreases their diffraction efficiency. The optimal energy to use for each write depends on the number of writes before and after it in that zone; and also on the number of reads we want to support on the zone before doing a refresh operation.

Read erasure. Reads use the same laser light as writes but typically at much lower power levels. Thus they also slightly degrade all the superposed holograms in the zone, including the one currently being read. While the erasure from a single read is much lower than that of a write, the cumulative erasure of a large number of reads can be significant. As holograms get weaker due to read erasure, they require increasing amounts of energy to read. For a fixed laser power, this means an increased read latency and reduced IOPS. In practice, this means that we must periodically do a *read refresh* — copy all live data out of a zone that has served many reads into a fresh zone. While previous work on holography assumed a constant amount of energy per read, we have found that it is much more efficient to use a variable read profile with the energy per read increasing over time until a refresh is triggered.

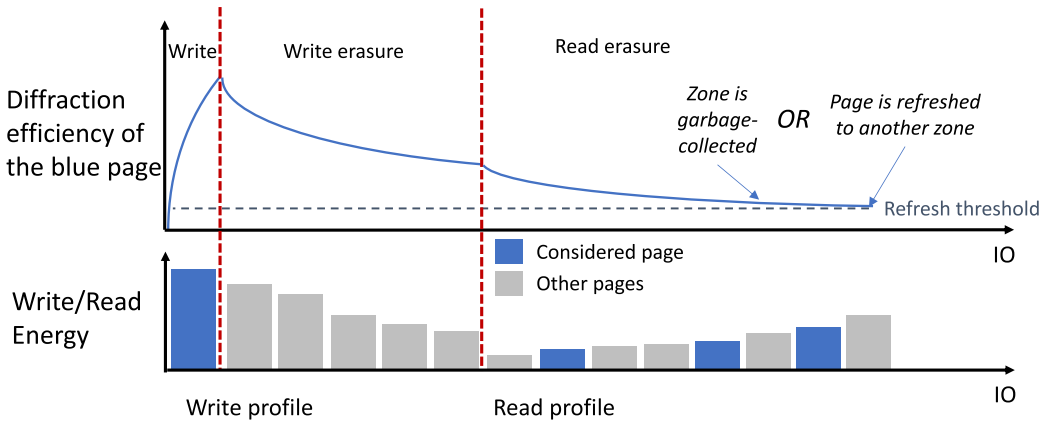


Fig. 9. Page (hologram) life cycle: evolution of the diffraction efficiency of a page in a zone over time.

Zone life cycle. We can think of each zone as having a life cycle from when the zone has just been erased until it again reaches a fully erased state. The lifecycle consists of a write phase in which holograms are written in an append-only manner, followed by a read phase where any page can be read at random at any time. For simplicity, we assume that reads will not interleave with writes. This is a valid assumption for a design where a small number of open zones is maintained for writing per device, and partially written zones are buffered in DRAM or SSD. Interleaving reads and writes would add some complexity but we do not believe it would significantly change the end-to-end energy efficiency of the system. Figure 9 shows the “life cycle” of a single hologram within a zone in a holographic storage system. The top of the figure shows the diffraction efficiency of a single page in the zone, starting from when it is written and progressing through writes of subsequent pages and the read phase. The diffraction efficiency of the hologram increases over time during the write as a function of the total write energy, but with diminishing returns. Reads to any hologram in the zone also cause erasure, but at a lower rate as reads typically use less energy. Eventually, the hologram must be refreshed by rewriting the data to a different zone. For write-heavy workloads, it is also possible that the entire zone is garbage-collected (giving us a free refresh of all live data in the zone) before the refresh threshold is reached. The bottom part of the figure shows the write profile with the page under consideration in blue and other pages in grey. Note that the shape of the read profile is independent of the actual pages being read as read erasure applies uniformly across the whole zone.

3.2 Holographic Translation Layer

Any practical holographic storage must be able to optimize the write and read profiles as well as handle the media management operations (garbage collection and read refresh) in a way that is transparent to the application. Figure 10 shows the design that we will use as our system model throughout this article. It includes a **Holographic Translation Layer (HTL)** (Figure 10(a)) that provides the abstraction of a rewritable, flat address space on a medium that needs variable write and read profiles as well as garbage collection and read refresh. This is similar to the notion of a **Flash Translation Layer (FTL)** in flash SSDs [1, 19, 20, 31, 44, 47, 66].

The HTL manages the life cycle of each page in each zone in a device. It tracks the history of each zone and makes the key decisions on how much energy to use for each write and read; when to refresh a page; and when to trigger garbage collection. We use a set of simple, parametrized heuristics to make these decisions. Our approach is then to use a combination of measurement, simulation, and modeling to find optimal values of these parameters for any given workload.

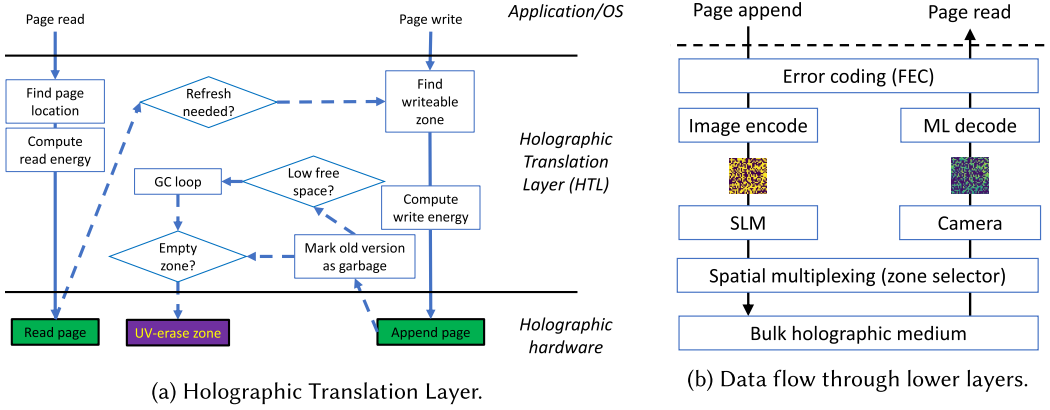


Fig. 10. High-level design of holographic storage.

Read profiles and write profiles can be computed on a per-zone basis, but for simplicity here we assume that they are computed offline as a single read and write profile for the entire system. The online read/write energy computation is then a simple index lookup onto the profile. Only a small number of zones is kept active for write (i.e., append) at any given time. This allows us to buffer all writes until we have an entire zone's worth of data, and only then to write the zones. This also means that we do not suffer read erasure during the write phase.

Refresh decisions are based on tracking the diffraction efficiency of reads, and rewriting the data when this goes below a threshold. In our simplified system model we precompute a fixed *read budget* per zone. We then maintain the invariant that the read budget of a zone always exceeds the number of live pages in it; i.e., each live pages can be read at least once more. If this threshold is reached then each subsequent read from the zone will trigger a refresh, causing the number of live pages in the read zone to decrease. When the number of live pages reaches zero, the zone is scheduled for UV-erasure.

Application writes and refresh writes both create garbage, i.e., the old version of the data that is being overwritten. GC frequency is governed by the amount of free space reserved for garbage collection; when the free space reaches this threshold, the zone with the fewest live pages is selected and the live pages are copied to another zone.

We keep only a few zones open for writing at any time, and the rest are in a read-only (or empty) state. We assume that the writeable zones are cached in a higher storage tier (e.g., DRAM or flash SSD) and flushed in their entirety when they are full: this means zones transition from append-only to read-only without interleaving reads and writes.

Many variations on these heuristics are possible, and there are also interesting low-level design choices, e.g., such as the division of HTL functionality between firmware and system software. We believe it is premature to optimize these details before the overall system is shown to be feasible for energy efficiency. We use these simple heuristics and an idealized system (e.g., we ignore concurrency and assume all operations are done sequentially) to explore the parameter space for read/write energy, read refresh, and garbage collection in simulation. Regardless of implementation details, the key challenge for the HTL is to balance the competing demands of application reads and writes with the background media maintenance operations.

There are several challenges to be solved before this architecture can be used in a practical system. First, we need to achieve sufficiently high density and energy efficiency to be competitive

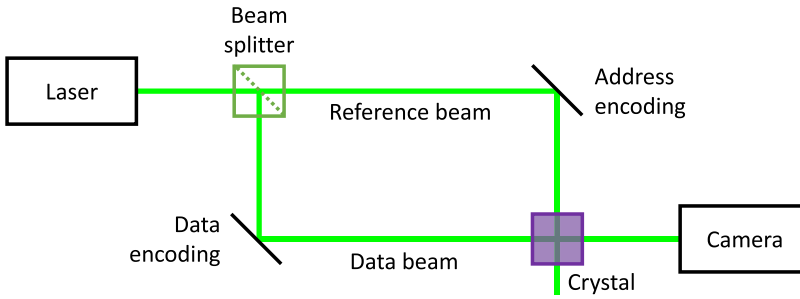


Fig. 11. Simplified schematic of holographic storage setup.

with other storage technologies. We have significantly advanced the state-of-the-art in both of these areas, as we will discuss in Section 4. Second, we need to solve the spatial multiplexing problem without using mechanical movement. Our spatial multiplexing design and the challenges associated with it are discussed in Section 5. Third, we need to solve the problem of reading noisy and distorted holograms at high density; our spatial multiplexing design based on waveguides adds additional distortions and complexity. The details of the ML algorithms used to compensate for these distortions are beyond the scope of this article, and are described in a separate research article [50].

3.3 Hardware Testbed

Alongside our simulation model we have built a hardware testbed to validate our simulation results. The testbed contains all the hardware components needed to write and read holograms within a zone and is fully automated. Testbed automation is controlled through software on a standard PC server, which also runs the **Forward Error Correction (FEC)**, image encode, and ML decode algorithms shown in Figure 10(b). The testbed allows us to run repeatable experiments with arbitrary read and write profiles and data, and to verify that data can be written, read, and decoded successfully. It also lets us easily experiment with different crystal samples, as well as different optical designs and components. Our testbed lets us focus on the main challenges such as the fundamental material properties as well as machine learning based decoding, while emulating/simulating the other parts of the system.

Figure 11 shows a simplified schematic of our testbed. A 532nm laser emits a laser beam which is split into a reference and data beam. The data is encoded on the *data beam* using a SLM as shown in Figure 12. The reference beam angle (i.e., the “address”) is changed using a set of mirrors and lenses (not shown in the figure). The two beams are directed to intersect within the crystal. To read the hologram, only the reference beam is required, and the diffracted readout beam is captured by a camera. Figure 13 shows an example of a camera image representing a page of data, read from the crystal in one of our experiments. It includes typical artifacts found when using commodity optical components, including noise and distortion.

Figure 14 shows a picture of the testbed used for our experiments. This is a typical setup and similar to one used by previous systems. This is still a testbed and not an end-to-end storage device — e.g., our read/write testbed does not include spatial multiplexing across zones which is needed for a capacity device; the HTL and other key components are implemented in software rather than firmware; and the components are not miniaturized and integrated as would be the case for a commercial device. However, it does not make sense to build such a device until the key challenges of energy efficiency, sensitivity, and spatial multiplexing have been solved. A testbed such as the

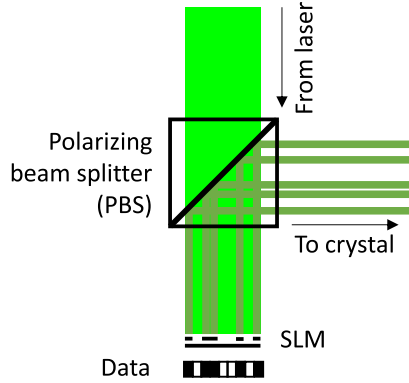


Fig. 12. Encoding data using an SLM.

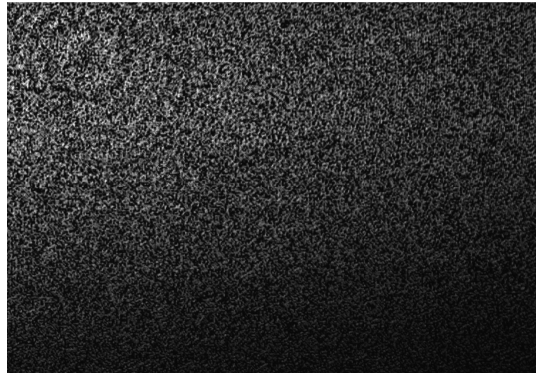


Fig. 13. Image of the hologram captured by the camera.

one shown gives us the flexibility to experiment with different crystals, optical components, and configurations.

4 Energy Efficiency

By writing and reading entire images at a time, holographic storage exploits the inherent parallelism of SLMs and cameras to achieve high data rates. At the same time, techniques such as angular multiplexing allow for rapid addressing by only steering optical beams. In principle this should lead to very high IOPS, limited only by the frame rates of the optical devices, e.g., cameras.

We find however that holographic storage IOPS are limited by the end-to-end *energy efficiency*: the amount of energy required per IO, including the amortized cost of read refresh and garbage collection and accounting for losses in the optical system. Typical hologram diffraction efficiencies are around 10^{-4} when 100 holograms are superposed, i.e., only 0.01% of the light from the reference beam is received at the camera. In addition, there are non-trivial energy losses at multiple stages of the process. The laser, the SLMs, the spatial multiplexing, and the crystal all contribute to the overall energy efficiency of the system. With standard components available today, the energy efficiency of the system is around 0.65 IOPS/W (Input Output per Second per Watt). By comparison a hard drive has an energy efficiency of 20 to 30 IOPS/W, while SSDs are around three orders of magnitude more energy efficient. Our estimate ignores the power consumption of the camera, the

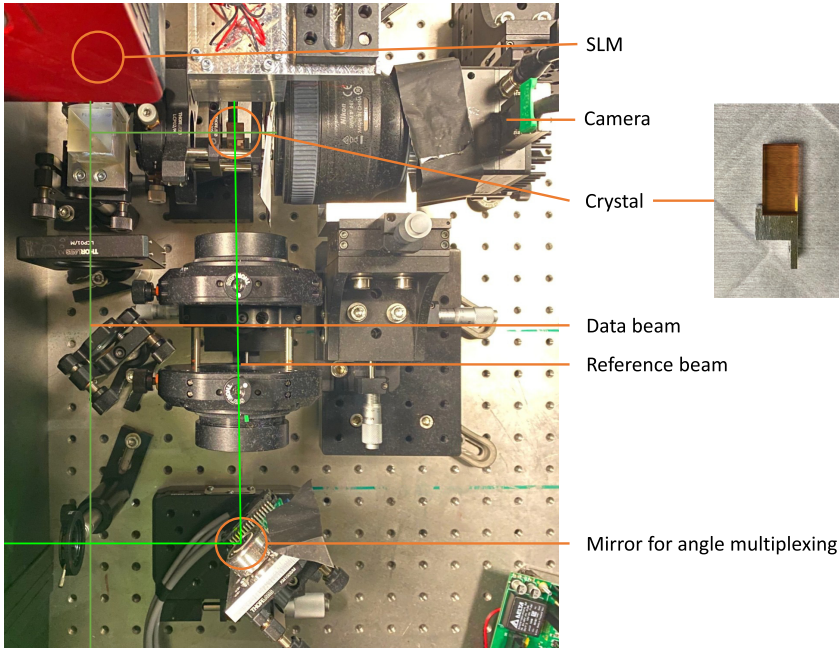


Fig. 14. Overhead view of hardware testbench.

SLM, and image decoding — the power consumed by the laser dominates the overall power and needs to be substantially reduced before we consider optimizing these other factors. It also ignores the power lost to spatial multiplexing — this is covered in Section 5.

A straightforward way to increase the IOPS would be to drastically reduce the density of holograms per zone. This would increase the diffraction efficiency of each hologram and reduce erasure, but would also reduce the density thus increasing cost per GB. Another simple way to increase the IOPS would be to increase the power of the laser. However, this would increase the cost of the laser, which is the dominant cost factor in the system. There is thus a need to maximize the energy efficiency of the system without reducing density.

We believe laser efficiency can improve from 2% to around 15% and that the cost per Watt of laser power will also drop when scaled to high-volume production. SLM efficiency also has some room for improvement. Read energy could be decreased by increasing camera sensitivity but this is likely to trade off signal-to-noise ratio (and hence effective capacity) for performance. Overall, for holographic storage to be competitive, we still need a significant reduction in the HTL overhead (garbage collection and read refresh) as well as the energy efficiency at the crystal (the average energy used per read and per write).

4.1 Density

The challenge with holographic storage is to achieve high end-to-end energy efficiency as well as high density. The best previous density achieved (by IBM) was 5.8 GB/cm^3 [15], with 1,000 pages using precisely aligned custom optical components. This density is sufficient to consider holographic storage seriously as a warm storage technology; however it was not clear if it was realistic for an end-to-end scenario and whether we could leverage commodity cameras and SLMs. Additionally reading back the pages was only validated only for 100 reads from the zone: we

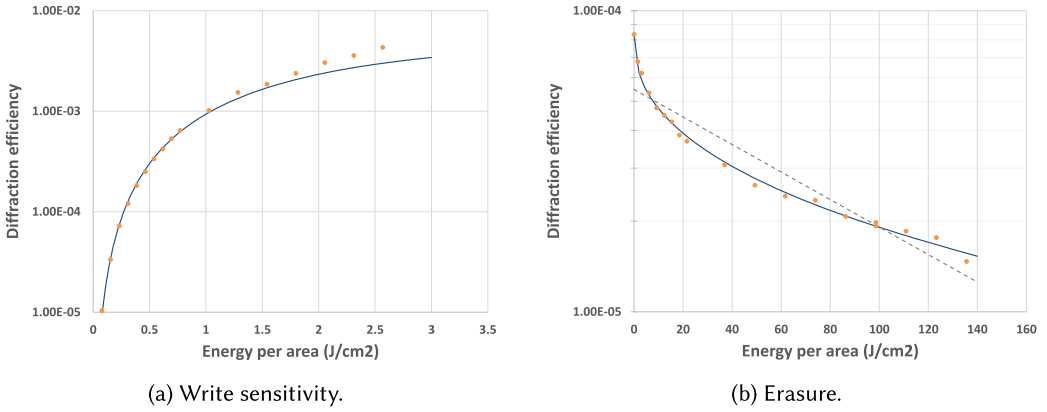


Fig. 15. Model fit for write sensitivity and erasure (reproduced from [23]). Dots are experimental data, solid lines are model fits, and the dashed line is the previous state-of-the-art model.

know that due to read erasure reads later in the zone lifecycle are more likely to fail, and for most workloads the zone lifecycle needs to extend well beyond only reading 10% of written pages.

Thus we asked the question *Can we achieve a competitive density, using commodity components, while allowing for read refresh and garbage collection, and would this be energy efficient?*. On the density front we were able to show a maximum density² of 9.6 GB/cm^3 using 705 pages/zone. This $1.8\times$ improvement over the previous state-of-the-art was enabled by using a denser encoding of bits onto symbols and optimizing the optical setup to shrink the zone size. In order to decode the data despite the increased density and the errors introduced by commodity cameras and SLMs, we used machine learning based decode. We also validated our results by checking that we could reliably read the zone sufficient times before read refresh was needed in our optimized configuration — the validation results are shown later in this section.

These densities are somewhat lower than that achievable with HDDs and significantly lower than that of SSDs. However, the bulk medium for holography is simpler and cheaper to manufacture than HDDs or SSDs. At the densities we have achieved, we believe that density is no longer the main obstacle to cost-effective storage. At these densities, and using commodity optical components, the overall system cost is already dominated by the laser cost, which scales with the amount of optical power needed. Energy efficiency now becomes the biggest challenge: can we significantly improve energy efficiency without sacrificing too much density or capacity? A key observation here is that the density that matters is user-visible density, which is proportional to the number of user pages net of error correction and also net of any free space set aside for garbage collection. As we will see, for many workloads it is more energy-efficient to achieve a given user capacity by overprovisioning the raw capacity. i.e., while a higher raw density does have a penalty in diffraction efficiency, it is more than compensated for by less frequent garbage collection.

4.2 Modeling the Medium

Figure 15(a) shows the measured *write sensitivity* — the diffraction efficiency of a single hologram in one of our sample crystals as a function of the write energy. We see that the diffraction efficiency increases with the write energy, but with diminishing returns as the medium begins to saturate. In practice we would operate in the initial region of this curve where the diffraction efficiency

²For a fair comparison, we report density net of error correction similar to the IBM result.

is increasing rapidly with the write energy. Operating near the saturation limit is not energy-efficient; it also means that fewer holograms can be superposed in the same zone due to write erasure, which reduces the overall capacity of the system. We see that in the region of interest, our experimental data (red dots) is a close match for the theoretically predicted curve (solid blue line). Figure 15(b) shows an *erasure curve* — the diffraction efficiency of a previously written hologram as a function of the energy used for subsequent writes or reads in the same zone. We found that existing mathematical models of erasure (dashed line) did not fit our experimental data, and we developed a new model, described in a companion publication [23], that takes into account the dynamics of the crystal and has a good fit to our experimental data. Based on our experimentally validated models of write sensitivity and erasures, given any sequence of writes and reads (each represented as an amount of optical energy received at the crystal) starting from a “reset” state we can compute the resulting diffraction efficiency of each hologram. This can be used to compute, for example, the point at which read refresh must occur in the lifecycle shown in Figure 9.

4.3 Write and Read Profile Optimization

The “life cycle” of a zone (Figure 9), can be divided into a write (append) phase during which new holograms are added to the zone, and a read phase during which the holograms are read. For each zone lifecycle, we can choose the length of the write phase (i.e., the number of raw pages per zone) and the length of the read phase (the number of reads that must be supported on a zone before the data is rewritten). In addition, we can tune the amount of energy used for each write and each read in these two phases. The optimal values of these profile parameters are a function of the crystal properties (sensitivity and erasure) as well as the workload properties. We map this to an optimization problem where the inputs are the desired user capacity in pages per zone, and the workload. The outputs are the optimal write profile and read profile. The length of the write profile is the number of raw pages per zone; typically this is larger than the target user capacity per zone to allow for garbage collection. The length of the read profile determines the point at which the zone must be refreshed to avoid read erasure.

There is a complex tradeoff between the HTL parameters and the read and write profiles, and energy efficiency. The end-to-end efficiency depends on the amount of free space reserved for garbage collection, the frequency of read refresh, the frequency of garbage collection, the energy per read, and the energy per write. Our first challenge therefore was to model the energy consumption of the entire zone lifecycle as a function of the write and read profiles and the workload. To do this we derived an analytic model of garbage collection and read refresh frequency, and combined it with the read/write sensitivity and erasure equations. Our analytic model assumes a uniform random read/write workload that is parametrized by the read/write ratio.

A second challenge is that the search space of write and read profiles is extremely large, making it infeasible for most optimization search techniques. From our analytic model we were able to express the energy consumption of the entire zone lifecycle as a function of just three parameters: the energy of the first write; the number of writes (i.e., the number of raw pages per zone); and the number of reads of the zone before a read refresh (the “read budget”). Table 1 shows the qualitative effect of each of these parameters on the HTL overhead and read/write energy at the crystal. This made it possible to use a standard genetic algorithm, NSGA-II [72] to search for optimal points. However, it was still slow due to the high cost of evaluating a single point, which took about 0.3s. By reformulating the problem to eliminate expensive root-finding inner loops and porting the implementation from Python to C++, we improved this by four orders of magnitude, to 0.3ms per evaluation. This made it feasible to find optimized solutions in about 15s–1min on a 20-core desktop machine. This means we can sweep the space of workload parameters, different media, and other system parameters rapidly while using the per-zone energy optimization as an inner loop.

Table 1. Parameters and their Impact

Parameter	Effect
Increase raw pages per zone (for a fixed amount of user data)	Decreases garbage collection overhead, increases refresh overhead, increases write energy and read energy per page
Increase write energy per page	Decreases read energy per page
Increase read budget	Increases read energy, decreases refresh overhead

4.4 End-to-end Optimization

End-to-end optimization of read and write profiles (i.e., including HTL overheads in the optimization loop) significantly improves the energy efficiency of the system. We compare this approach to the previous state of the art state-of-the-art [57] where the write profile was chosen to achieve equal diffraction efficiency for all holograms at the end of the write phase, chosen to be 60% of the maximum achievable diffraction efficiency for the same number of superposed holograms on a given crystal. Intuitively this approach is aiming for the “knee” of the diminishing returns curve in Figure 15(a).

Our approach improves on the previous one in three ways. First, we use *minimal reads* that use only the minimum energy required at the current diffraction efficiency of holograms in the zone. This significantly extends the lifetime of the zone before read refresh is required. Second we *jointly optimize* the write and read profiles using our improved media erasure model. Finally, we use our analytic model of GC/refresh to do *end-to-end optimization* that includes the overhead of these media management operations. This lets us optimize an additional key parameter — the number of raw pages per zone. A larger number of raw pages reduces the frequency of GC (as we have more free space to absorb writes), but increases the impact of erasure as well as the overhead of copying zones for GC and read refresh.

We evaluate the contribution of each of these improvements by comparing four configurations. We choose as a baseline a target user page capacity of 360 pages/zone, which is 50% of our maximum raw density — this allows us to explore overprovisioning as a way to reduce GC overhead. We use a uniform random workload with a read/write ratio varying from 0% to 100%. We compare the following configurations:

- Baseline: write profile with a fixed (60%) diffraction efficiency target and flat read profile, and a fixed allocation of 10% free space for GC (i.e., 400 raw pages per zone).
- Minimal reads: same as the baseline, but using minimal reads.
- Jointly optimized: write and read profiles jointly optimized using our new method, but raw pages per zone still fixed.
- End-to-end optimized: write and read profiles and the number of raw pages per zone were all jointly optimized.

Figure 16(a) shows the estimated energy efficiency of the system for the different methods, as a function of the read/write ratio of the workload. While the actual efficiency at the crystal is much higher — the overall numbers are scaled by the low efficiency of the laser (2%) and the SLM (25%). We see that end-to-end optimization significantly improves the energy efficiency. At 50% reads the efficiency is 14× higher than the baseline. However, the overall efficiency is still extremely low (0.65 IOPS/W) even with end-to-end optimization, and significant improvement is needed to make this competitive with HDDs. Figure 16(b) shows that when the HTL overhead (refresh and GC) is subtracted out (dashed lines), the jointly optimized and end-to-end optimized configurations perform very similarly. However, when HTL overhead is also considered (solid lines), the end-to-end optimized configuration is significantly more efficient.

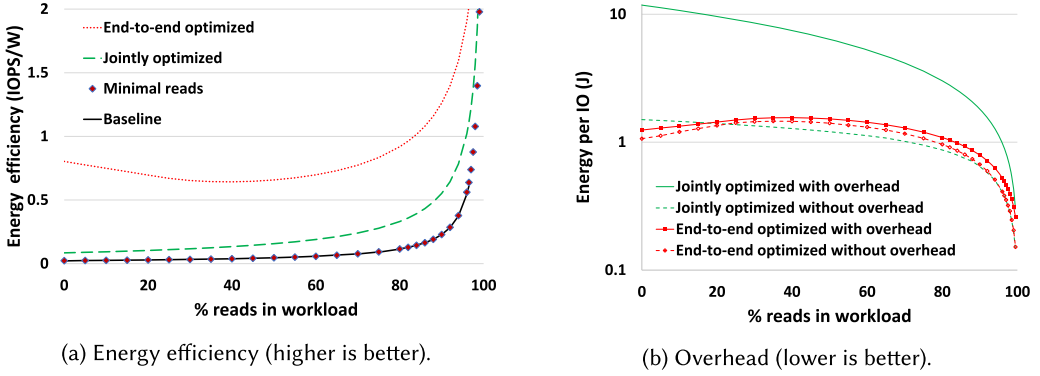


Fig. 16. Energy efficiency and overhead with different optimization methods for a uniform random workload.

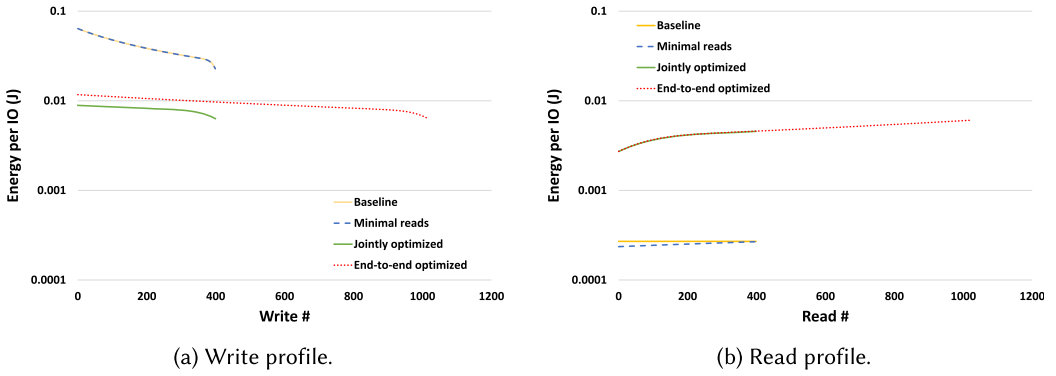


Fig. 17. Profiles generated by different optimization methods for a uniform random workload with 50% reads.

For extremely read dominated workloads (with more than 99% reads) the efficiency increases sharply regardless of method. Intuitively this is because there is almost no overhead for garbage collection, and the approach of using high energy for writes works well: writes are rare, and stronger writes means less frequent read refresh.

To illustrate the tradeoffs involved in optimizing write and read profiles, Figure 17 shows the profiles generated by the different optimization methods for a 50% read workload. The baseline approach is not workload- or GC-aware. For this workload, it uses too much energy for writes and the default space allocated for GC is too low. Using minimal reads slightly reduces the read energy; however the read energy is already very low and the overall energy usage is dominated by writes with these profiles. Joint optimization of writes and reads achieves a better ratio of read and write energy resulting in a 3× improvement in overall efficiency. However, the efficiency is still limited by the need for frequent garbage collection.

Full end-to-end optimization finds a very different point in the space with much higher efficiency, by increasing the number of raw pages per zone by 2.5×, from 400 to 1,018 (the x-axis on Figure 17(a) shows the length of the write profile which is also the number of raw pages per zone). This reduces the GC overhead enormously (see Figure 16(b)) while slightly increasing the write and read erasure; the net effect is a further improvement in efficiency of almost 5×. While the benefits of reserving additional space for GC are similar to that of overprovisioning capacity

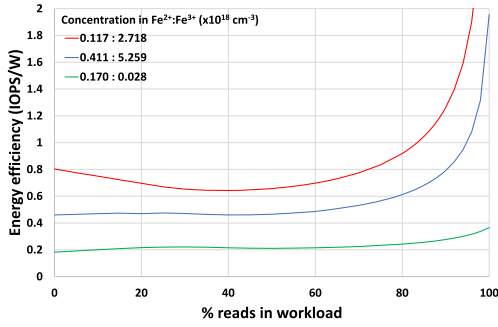


Fig. 18. Effect of different crystal dopant concentrations and different read/write ratios.

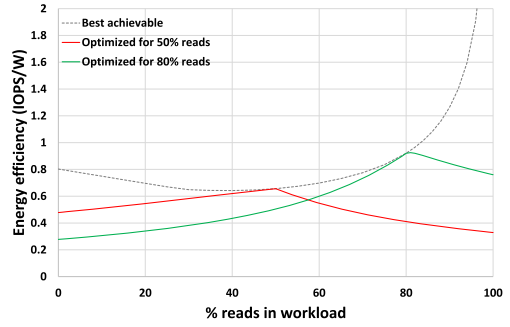


Fig. 19. Optimizing for workload read/write ratio.

in flash SSDs or other zoned storage, it is important to note that this is done without sacrificing overall density and hence without increasing cost. We are exploiting the non-linear relationship between density and energy efficiency to achieve a higher overall efficiency for a given user-visible capacity. This gives us the freedom to overprovision capacity by a large factor if this is the best choice for overall efficiency. The optimized write and read profiles obtained with this approach have been validated on our experimental test bed [23], to ensure that pages can be reliably read and written using these profiles.

4.5 Read Validation

While diffraction efficiency is the key metric for measuring hologram strength, for a storage system we need to ensure not just that holograms are sufficiently strong, but also that the data can be read back reliably at a given level of error correction. We have validated this experimentally using the write and read profiles generated from end-to-end optimization, for both 50% reads and 100% reads. In both cases, we were able to read back pages up to the end of the generated read profile (i.e., up to the point of read refresh) with a BER (bit error rate) that was correctable with the configured amount of error correction. Further details on the validation experiments are in our companion publication [23]. While a fully deployed system would require much more validation — this gives us confidence that our profile optimization method gives feasible configurations.

4.6 Media Optimization

Our lab testbench for characterizing the medium, our media and workload models, and our optimization methods have already helped us to pick the best performers from a small set of candidate crystals (the results shown so far are based on this best crystal). Figure 18 shows the optimized energy-efficiency of three of these crystals. These are all based on iron-doped Lithium Niobate (Section 3), with different concentrations of the dopant ions (Fe^{2+} and Fe^{3+}). The dopant concentrations affect the media properties: e.g., the write sensitivity first increases and then decreases as Fe^{2+} concentration is increased. Further details of the media characterization are in our companion publication [23].

4.7 Workload-specific Optimization

Write/read energy profiles and HTL parameters are runtime parameters that can be tuned dynamically for the workload. Figure 19 shows the effect of optimizing for the precise read/write mix of the workload (dotted line), for a uniform random access pattern. The solid lines show the performance when optimized for a fixed read percentage of 50% (red line) or 80% (green line) —

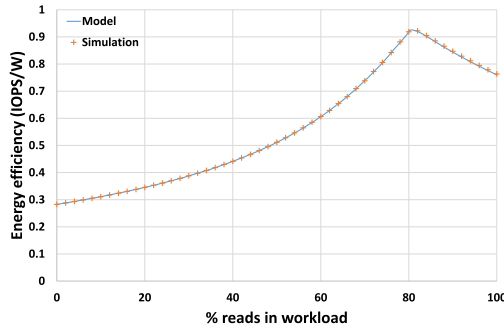


Fig. 20. Model fit for uniform random workloads.

as the actual workload moves further away from the optimization target, the penalty for using a workload-independent configuration increases significantly. Cloud storage providers have a good understanding of their workload, and such workload-specific tuning is feasible in practice. The main reason this is not possible in today's cloud storage hardware is legacy interfaces that offer very limited scope for tuning low-level hardware parameters for the high-level workload. We believe that enabling dynamic, full-stack tuning down to low-level hardware parameters is an important principle not only for cloud holographic storage but also for cloud-first design in general.

The results shown in this section are based on analytic models of the workload, that assume a uniform random access pattern with a fixed read/write ratio. This is very useful for exploring the design space, and uniform random access patterns capture the worst-case scenario that we can expect to see for energy efficiency. We are also able to use the same optimization framework with a trace-driven simulator that can compute energy usage given a trace of read and write operations at the block level. We have cross-validated our analytic model against the simulator comparing the computed energy efficiency for uniform random access patterns. Figure 20 shows the fit between the model and a simulator for a uniform random workload, using a profile optimized for 80% reads. More details about the model fitting are in a companion publication [23].

We used the simulator to explore other workloads, e.g., skewed access patterns as well as sequential access. While the energy efficiency of the system does vary across these workloads, the overall trends are similar to those observed for uniform random workloads.

4.8 Improving Efficiency Further

Is a $14\times$ improvement over the state-of-the-art sufficient to make holographic storage competitive? Unfortunately it does not, even with optimistic assumptions on technology improvements. We assume that laser efficiency would improve from 2% to 15% and also that laser costs will drop significantly with volume production. Based on input from laser vendors, our estimate for laser cost at volume is \$400 per optical Watt of output power (or \$60 per electrical Watt at 15% efficiency). We also assume that the other components (cameras, SLMs, crystal) contribute negligibly to optical power loss, electrical power consumption, and dollar cost. This is very optimistic, especially in the case of spatial multiplexing, described in the next section. Even under these assumptions, our analysis shows that we need at least another $13\times$ improvement in end-to-end energy efficiency to be cost-competitive with HDDs. In reality, given the non-zero contribution of other components to cost and power and the challenges of spatial multiplexing, we need an even larger improvement.

In our view, this can only be achieved by fundamental improvements in the physical medium. In other words, despite the attractive properties of iron-doped Lithium Niobate, it is unlikely to be the way forwards for holographic storage. At minimum we need new materials; possibly we need

to exploit new physical effects beyond the electro-optical properties that are used for holography in Lithium Niobate.

5 Spatial Multiplexing

A naive implementation of holographic storage would allocate one camera and one SLM to each zone of the medium. However, this is not practical from a cost point of view. While the bulk medium is cheap, the cost of the camera and SLM is non-negligible and needs to be amortized over tens of terabytes of capacity. On the other hand, several factors limit the capacity of a single zone to tens of megabytes. Write erasure limits the number of pages that can be written to a single zone, and the SLM and camera resolution limits the size of each page. These limitations cannot be addressed simply by using larger zones. Instead, we must scale by multiplexing a single camera-SLM pair over a large number of relatively small zones spread over the bulk medium. Each zone would be a few millimeters in dimension, and we would need thousands to millions of such zones per camera-SLM pair. We refer to this multiplexing as *spatial multiplexing* [16] as distinct from multiplexing within a zone, e.g., angular multiplexing.

Previous approaches assumed the crystal would move with respect to the optics to allow access to different zones [5, 60]. e.g., by using a circular “platter” like design which would both rotate and translate. This approach is not practical for high IOPS workloads, as the mechanical movement would limit the number of IOPS that could be achieved. HDDs — which are heavily optimized for rotational and seek latency — are already struggling to keep up with the IOPS demands of cloud storage. Holographic crystal is much heavier than HDD media making its movement at speed (while limiting vibrations and forces exerted on the medium) much more challenging than HDDs’ platters. The alternative — moving the optical components such as cameras and SLMs with respect to a fixed medium — is also impractical. The cost advantages of using commodity optical components come with the disadvantage of relatively large size and weight. The only way to achieve high IOPS is to move light rather than the mass, i.e., steer the optical beams to the correct zones.

We considered a design based on MEMS mirrors and a single scan lens [34]. However, using available lenses this design is fundamentally limited to a few tens of zones [16]. We decided instead to rethink spatial multiplexing from scratch, and converged on an “optical imaging network” that uses waveguides to transport entire images along a straight line path. It uses binary switching elements composed of pi-cells [10] and polarized beam splitters [40, 82]. Each switching element can either transmit the image along the waveguide, or reflect it at right angles to the original direction. Using these elements, we can imagine networks that transport SLM images from a single SLM to one of many zones; and conversely from one of many zones to a single camera.

Waveguide propagation is almost lossless but the switching elements introduce losses. Even a small loss per element can add up to a large loss if the image must propagate through many elements; thus the number of switching elements on the path is a key metric. We investigated multiple network designs with different tradeoffs on cost/complexity, physical volume, and optical power loss.

A binary tree configuration (Figure 21(a)) is optimal in terms of losses. However, it is complex to manufacture and crucially it has an extremely large physical footprint. To multiplex over 1 M zones (about 28 TB with our current optical components) requires three trees each of depth 20: one each for the data beam, the address beam, and the readout beam. As imaging waveguides (unlike fibers) can only propagate light in a straight line, and the cross-section of the waveguide is equal to the width of the zone, this means that the volume of the waveguide network is 120 times that of the crystal it addresses. This is a loss of 120× in effective density, making the design unattractive for rack density which is a key metric for cloud storage.

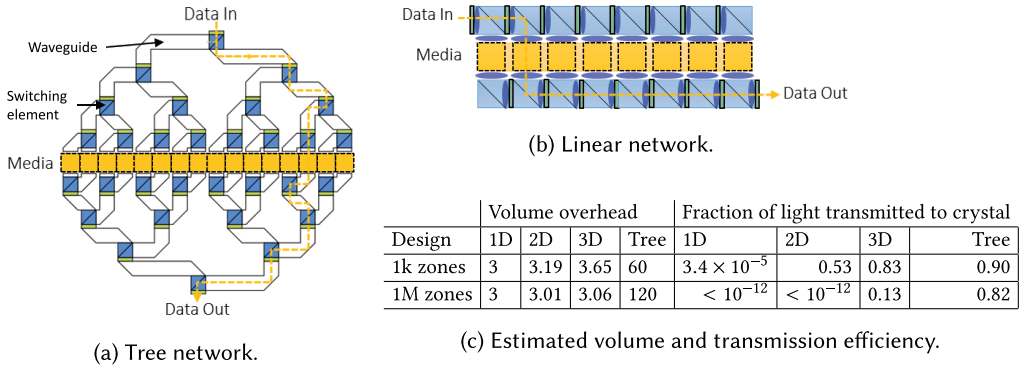


Fig. 21. Possible networks for spatial multiplexing with estimated volume and losses.

At the other extreme, a linear design (Figure 21(b)) has good density but has extremely high losses due to the large number of lossy switching elements in each path. Extending the linear design to 2D or 3D reduces the losses significantly with little increase in volume. However it increases complexity and cost, and the loss is still high at the scale of 1 M zones. Table 21(c) shows the estimated volume overhead and optical power transmitted for different configurations. This assumes 1% loss per switching element which is somewhat optimistic — the elements available today have losses of 2%–3%.

A second challenge is the need to correct for complex distortions introduced by the waveguide switching network. We have had partial success in using machine learning to correct for these effects [50]; the method works well for ideal waveguides but is very sensitive to manufacturing defects that cause slight deviations from the ideal rectangular cross-section.

6 Related Work

Holography was first described in 1948 by Gabor [35, 36, 37]. In the 1960s, the concept of Bragg selectivity emerged [45, 49, 75], enabling precise readout of a single hologram from a volume containing multiple superimposed holograms. This ability to store multiple holograms in the same volume made holography attractive for data storage, not just for images.

The theory, physics, and optics of holographic storage have an extensive literature. There have been a few end-to-end storage prototypes and no commercially successful products. Our aim in this section is not to provide a comprehensive review of the literature, but to highlight the key works that have influenced our research and to provide context for our contributions. For more detail, we recommend the book *Holographic Data Storage* [25] as an excellent introduction to the history and fundamentals of the field as well as a comprehensive review of the literature up to the year 2000.

6.1 Rewritable Holographic Storage

Early attempts at high-density rewritable holographic storage were severely limited by the available hardware components. One of the earliest uses of angle multiplexing from the 70s [33] was limited to 10 pages of 10 Kb each. Other prototypes from the 70s [54] showed the ability to erase and rewrite the crystal without degrading it.

The most complete and highest density rewritable holographic data storage prototypes we are aware of are the IBM DEMON AND DEMON 2 [14, 15, 68], designed in the late 90s. This system used an SLM with 1024×1024 pixels and matching cameras with very precise alignment to store

large holograms. They relied on an error correction code that encoded 8 bits in 12 pixels, resulting in a page size of 670,816 user bits. By angle-multiplexing 1,000 holograms in a single zone of $(1.6 \times 1.6 \times 5.5 \text{ mm})$, they demonstrated a net density after error correction of 44 Gb/cm^3 . At these densities it finally became feasible to think about holographic storage as a competitor to HDDs. However, it was still limited by the high cost of the custom optics; the need for precise alignment; the need to compete with HDDs in their Moore's Law phase; and the very low energy efficiency. The system was never commercialized.

6.2 WORM Holographic Storage

Holographic storage has been more widely studied and prototyped for **write-once, read-many (WORM)** applications such as archival storage. These typically use photosensitive polymer media, often in a relatively thin film coated on a spinning substrate, i.e., an optical disk form factor. The media manufacture is even cheaper than for rewritable media, and the energy costs for read are lower. The use of mechanical movement (i.e., rotation and seek) limits IOPS, but this is less of an issue for archival storage than for warm storage. IOPS requirements are typically lower for archival storage (the competition is magnetic tape), and latency is often dominated by other factors (such as retrieval from a library).

Several such systems [51, 53, 63, 74] have been proposed to compete with archival media such as optical DVDs but few have been commercialized. One of the earliest attempts in this space is from the 60s [51] which wrote 4 sets of 4,096 holograms, each with 32×32 bits that could be read by a single device. This would result in a storage capacity of 2 MB per disk with a reported bandwidth of 16 MB/s. The density was relatively low as angle multiplexing had not been developed yet.

Holoscan [74] (1974) is, as far as we know, the only holographic data storage system to reach commercialization. It was a read only sequential memory intended for checks verification. It had a capacity of 1.5 MB and a read bandwidth of 112 Kb/s. In the late 90s Tamarack Multi-store [63] and others [53] proposed systems with capacities in the 500 MB – 1 GB range per disk. More recently InPhase [7, 26] demonstrated the storage of 300 GB in a single disk (CD-ROM shaped, 1.5 mm thick medium) and reached a throughput of 20 MB/s. However, this did not succeed commercially and currently, there is no commercially available holographic storage technology.

Write-only holographic storage for archival applications continues to be an active area of research. Current proposals aim for an optical disk form factor with a capacity of 1 TB and a throughput of 1-10 Gb/s [42]. However the capacity is still not high enough to be competitive, and the lifetime of the polymer materials used for WORM holographic storage is still a concern, compared to other more durable materials, e.g., glass [3, 4].

6.3 Photorefractive Materials

The atomic changes happening during hologram formation have been modeled by Kukhtarev et al. [46]. This model describes the movement of electrons in the material and is the foundation of metrics used to evaluate materials, including the sensitivity of a material or the erasure constant that describe how easy it is to record a hologram and how stable it is when read. Mok et al. [57] introduce $M\#$ (read “M number”), a characteristic of the material that links the number of holograms superimposed M to their diffraction efficiency $\eta = (M\#)^2/m^2$.

Photorefractive materials have been studied in details and the results have been compiled in books such as *Lithium Niobate: Defects, Photorefraction and Ferroelectric Switching* [76] for lithium niobate in particular or *Photorefractive Materials and Their Applications 2: Materials* [38] for a range of materials covering polymers as well as inorganic crystals.

The fact that Lithium Niobate has a tradeoff between write sensitivity and erasure is well known [41]. While end-to-end optimization can improve the tradeoff significantly as shown by our work, this tradeoff still exists and continues to be a challenge in making holographic storage competitive with other storage technologies.

6.4 Beam Configuration

There have been several attempts to optimize holographic storage by optimizing the optical beam properties. From a theoretical perspective, density is maximized when the two beams intersect at right angles to each other in the medium [11]. Alternatively using a “transmission geometry” where the two beams intersect at a very narrow angle, can significantly improve write sensitivity at the cost of much lower density. In our work, we primarily use the 90-degree configuration as our goal was to optimize energy *without losing density*. It would be interesting to extend our media model and end-to-end optimization to include transmission geometry as a design parameter.

Write efficiency for a small number of holograms can also be improved by using a higher intensity of laser beam [70] to exploit non-linearities in the hologram formation process. However, this effect is not significant when writing many hundreds of holograms in a zone. In practice, we already use the maximum intensity possible from the available laser and optics — even if this does not improve energy efficiency significantly, it reduces the time to deliver the same amount of energy and hence improves IOPS.

6.5 Phase Multiplexing

Angle multiplexing [17, 48, 49, 56, 73] is the most common method of superposing multiple holograms in the same zone, by changing the angle of the reference beam and exploiting Bragg selectivity [45, 49, 75] to minimize cross-talk between the different holograms.

Phase codes [8, 27, 29, 30, 58] are another method to superimpose multiple holograms in the same zone. The reference beam is modulated using an SLM. This reduces system complexity by avoiding a dependency on fast and precise MEMS mirrors — we can simply use SLMs, which are already being used for the data beam. In theory, orthogonal phase codes such as Walsh–Hadamard codes can have lower cross-talk and a better signal-to-noise ratio than angle multiplexing [27]. After extensive experimentation we found that in practice, commodity SLMs can not exactly implement ideal phase codes and this results in higher cross-talk and noise than angle multiplexing. We were able to write 500 pages at 47 KB each, slightly better than the previous state-of-the-art of 480×37.5 KB for phase multiplexing [29]. However, this is lower than we can write with angle multiplexing; we also saw higher erasure and much higher **bit error rate (BER)** on readout than with angle multiplexing. It is possible that with better SLMs, phase multiplexing could be competitive or even surpass angle multiplexing on density; we do not expect the energy efficiency to improve however.

6.6 Spatial Multiplexing

To the best of our knowledge, we are the first to explore mechanical movement free spatial multiplexing for holographic data storage, and the first to develop an optical waveguide based switching network for image transport. Certainly all previous attempts at spatial multiplexing for holographic storage relied on mechanical movement. DEMON and DEMON 2 [14, 15, 68] relied on movement of the medium to address multiple zones. Tamarack Multistore [63] also moved the optics radially (except the laser) in a manner similar to disk readers. Burr et al. [16] used a mechanical scanner to address 16 different volumes each storing 10,000 holograms.

6.7 Translation Layers and Garbage Collection

The FTL is a key component in flash-based storage systems. It is responsible for mapping logical addresses to physical addresses, garbage collection, and wear leveling. FTLs balance multiple objectives [1], including maximizing the lifetime [19, 20] of the flash memory, maximizing storage space [44], and improving performance [31, 47, 66]. Our HTL design is analogous to an FTL and occupies a similar place in the storage stack. However, the optimization parameters are different, in particular the value of the HTL is in jointly optimizing the media energy profiles with garbage collection. Unlike flash, read refresh is a crucial management operation, whereas wear leveling is not.

7 Discussion

The focus of this article has been on energy-efficiency and spatial multiplexing as these were (and still are) the main challenges to solve to make the technology practical as cloud storage. In the course of doing this research, we have also identified some interesting areas for future investigation, both in the context of holographic storage and in the broader context of storage systems.

Machine learning. We have used machine learning in Project HSD to compensate for the distortions and aberrations introduced by commodity optical components. In particular, ML helps us solve problems related to *alignment*, i.e., the mapping between image pixels and user data bits. Without alignment, the data is not decodable at all, even with large amounts of error correction. ML decode also lets us encode 2 bits per image symbol where conventional signal processing only lets us decode 1 bit per symbol.

The use of **convolutional neural nets (CNNs)** for processing images is well established, and as the fundamental unit of holographic read and write is the image, we are able to leverage the existing work on CNNs. We are also able to exploit recent and ongoing advances in machine learning hardware, especially inference-optimized hardware. While the use of ML on the data path would have been unthinkable a decade ago, today we have hardware and models that can decode data at high data rates and modest hardware cost and power. These models can correct for complex effects such as waveguide propagation that are extremely difficult to compensate with traditional physics-based approaches.

While other storage hardware will potentially have different sources of error and distortion, we believe this is a promising area of investigation for optical image based data storage in particular and for all kinds of storage in general.

Beyond image processing, we have also found machine learning, in particular Bayesian optimization [78] useful as a design tool. We have used Bayesian optimization to optimize hardware settings on the write path, specifically the intensity values to use on the SLM for each symbol value to be encoded. We have also used Bayesian optimization as an alternative to genetic algorithms for our end-to-end optimization of HTL and media parameters. Bayesian optimization is much more sample-efficient than the genetic approach, i.e., it evaluates far fewer points while searching for the optimal solution. However, it is also more computationally expensive per point generated. Overall the genetic algorithm was clearly the right choice when using our fast analytic model of the system — it converged an order of magnitude faster in real time than the Bayesian approach. However, we believe there is value in investigating the Bayesian approach in more depth, in particular when using expensive simulation models where the computational cost per sample is dominated by the simulation itself.

Full-stack optimization. So far we have focused on the challenges and opportunities for optimization below a block (or page) read/write storage interface. e.g., on jointly optimizing the HTL

parameters and the media parameters. However, we believe there is also significant potential for joint optimization across more layers of the storage stack. e.g., log-structured layouts such as LSM trees [22, 43, 61] are a natural fit for zoned storage. Many variants of LSM trees exist, offering a tradeoff between capacity, write amplification, and read amplification [6, 28, 52]. As similar tradeoffs exist at the HTL and at the crystal level, it would be interesting to consider if making the optimization even more “end-to-end” by including the application data structure, could improve the overall system performance.

Adaptive hardware. In Project HSD we started with a clean-slate design — to maximize the benefit of designing for the cloud free of legacy constraints. This has allowed us to optimize across layers in a way that would not be possible otherwise. e.g., there is no way to exploit workload knowledge to change the low-level parameters of a 3.5” HDD such as the amount of energy used per read or per write. With the HSD design we can adapt the low-level parameters such as the read/write energy profiles to the workload, and this has been crucial to the energy efficiency improvements. We could also adapt different zones within the same device differently, e.g., to support different workloads or different SLAs — we have not explored this direction but it is an interesting area for future research.

Tail latency. Latency distributions and tail latency are important metrics for warm storage. We have not considered them in our analysis so far — in our model we have a single powerful laser that is multiplexed over a large amount of data. The multiplexing can be done rapidly; and at full power the latency of individual reads and writes is small, even for high-energy operations (i.e., at the beginning of the write profile and the end of the read profile). It is likely that further down the line — if/when the challenges of energy efficiency and scalable spatial multiplexing are solved — we will need to revisit the issue of latency. In particular, if a large number of small lasers is more cost-effective than a single large multiplexed laser, then latency may prove to be a new challenge.

Optimizing for skewed workloads. The work we presented is focused around uniform random workloads. However, in practice, workloads are often skewed, which open up new opportunities for optimization as the device can be reconfigured for different workloads at runtime. For example, pages written could be classified according to their read and write patterns, and then written to different zones with different media parameters in order to maximize the efficiency of the device. While this is an interesting research area, we expect that at best, this would still be limited by the best achievable energy efficiency for a known read/write ratio (Figure 19), which by itself is too low to be competitive.

8 Conclusion

Rewritable holographic storage is in principle a great solution to the challenges faced by warm cloud storage. Previous attempts to commercialize this technology (until about 20 years ago) were limited by several factors. HDD capacity per dollar was still growing exponentially; cameras and SLMs were slow, expensive, and low-resolution; it is not possible to design practical holographic systems in consumer device form factors.

In Project HSD we have re-examined holographic storage as a cloud-first technology, leveraging commodity optical components, machine learning, and the freedom to redesign the storage stack from top to bottom. We have found that in today’s world the main challenges to solve for holographic storage are energy efficiency and spatial multiplexing. While we have made advances in both these areas, including a 14× improvement in end-to-end energy efficiency, these remain the primary challenges.

To make holographic storage practical, we need significant improvements in the medium, i.e., new materials that preserve all the advantages of iron-doped Lithium Niobate but with 1–2 orders

of magnitude higher energy efficiency. We also need a spatial multiplexing system that is scalable, low-loss, and compact. While there are other interesting research problems to be solved in the context of holographic storage, we believe that these two are the most critical to solve before revisiting the viability of holography as a storage system.

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